Project Title:

Towards a Serverless Intelligent Firewall: AI-Driven Security, and Zero-Trust Architectures

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Abstract

The ephemeral and stateless nature of serverless computing environments presents unprecedented challenges to traditional network security frameworks, particularly in implementing Zero-Trust Architecture (ZTA) principles. Conventional perimeter-based defenses and rule-based Intrusion Detection Systems (IDS) are fundamentally ill-equipped to protect transient, event-driven functions that lack persistent state and infrastructure. These limitations create critical security gaps in modern cloud-native deployments where functions may execute for milliseconds before terminating, making continuous monitoring and contextual threat assessment extremely difficult.

This research presents a comprehensive **Serverless Intelligent Firewall framework** that synergistically combines deep learning-based intrusion detection with Zero-Trust enforcement mechanisms to deliver adaptive, real-time threat detection and mitigation in cloud-native environments. Drawing on the CICIDS2017 benchmark dataset, the framework employs a **Long Short-Term Memory (LSTM)** neural network to capture complex temporal patterns and sequential dependencies in network traffic behavior, enabling the detection of sophisticated anomalies and coordinated attack sequences that evade stateless detection approaches.

To address the inherent bias present in class-imbalanced datasets, a strategic undersampling approach was designed and implemented to equalize class representation and prevent model bias towards majority classes. The proposed **3-layer LSTM architecture** (hidden units: 128→64→32, dropout rate: 0.3, Adam optimizer, learning rate: 0.001) achieved exceptional performance of **98% accuracy, precision, recall, and F1-score** across all evaluation metrics, significantly outperforming baseline models including Decision Trees (DT: 90.2%), Support Vector Machines (SVM: 88.4%), and Convolutional Neural Networks (CNN: 93.0%). Statistical significance of improvements was confirmed through paired t-tests ($p < 0.05$). The results validate the efficacy of the proposed architecture in delivering scalable, adaptive, and Zero-Trust-compliant security for modern serverless environments.

Keywords: Serverless computing, intelligent firewall, Zero-Trust Architecture, intrusion detection system, LSTM, deep learning, cybersecurity, CICIDS2017, real-time threat detection, cloud-native security.

CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

Cloud-native and serverless computing have fundamentally revolutionized computing paradigms in the contemporary era by enabling elastic scalability, resource efficiency, and rapid deployment cycles. These architectural innovations allow organizations to focus entirely on business logic while infrastructure management is abstracted away by cloud providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform. The serverless execution model — exemplified by AWS Lambda, Azure Functions, and Google Cloud Functions — has become increasingly central to modern application development, offering automatic scaling, pay-per-execution pricing, and rapid deployment capabilities.

The fundamental shift from monolithic, long-running server instances to ephemeral, event-driven functions introduces unique and profound security challenges. Serverless architectures are characterized by short-lived execution environments, shared multi-tenant infrastructure, limited runtime visibility, and inherently stateless function execution. These characteristics fundamentally undermine traditional security assumptions. Conventional firewalls and perimeter-based security mechanisms were designed for persistent, stateful environments where network boundaries are well-defined and traffic patterns are relatively stable. In serverless architectures, these assumptions break down completely.

Traditional Intrusion Detection Systems (IDS) rely on maintaining state across multiple requests, learning traffic patterns over time, and correlating events across sessions. In serverless environments, each function invocation is isolated, potentially running in a fresh container, making it nearly impossible for stateful detection systems to maintain the contextual awareness required for accurate threat identification. Furthermore, the transient nature of serverless functions means that traditional signature-based and rule-based approaches cannot adapt quickly enough to detect sophisticated, coordinated attacks that span multiple function invocations.

The proliferation of cyberattacks targeting cloud-native infrastructure has accelerated dramatically in recent years. Attack vectors such as Distributed Denial of Service (DDoS) attacks, injection attacks targeting API endpoints, data exfiltration through serverless functions, and lateral movement within cloud environments pose existential threats to organizations relying on serverless architectures. The CICIDS2017 dataset, developed by the Canadian Institute for Cybersecurity, provides a comprehensive benchmark encompassing over 2.8 million labeled network flows representing a diverse range of attack scenarios, making it an ideal foundation for developing and evaluating intrusion detection capabilities.

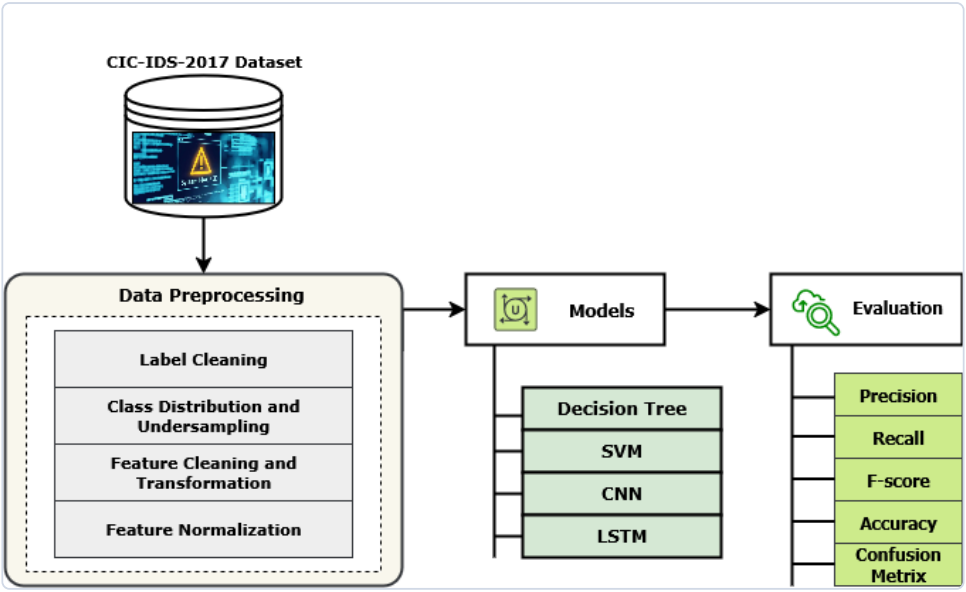

Figure 1. Graphical visualization of the overall research methodology — from CICIDS2017 dataset ingestion through preprocessing, LSTM model training, serverless deployment, and Zero-Trust enforcement.

1.2 Zero-Trust Security Paradigm

Zero-Trust security has emerged as a revolutionary security model that fundamentally challenges the traditional notion of implicit trust within network perimeters. The Zero-Trust principle, articulated in NIST Special Publication 800-207, can be summarized as "never trust, always verify" — a paradigm shift that assumes no user, device, or network segment is inherently trustworthy, even within organizational boundaries. This principle is particularly relevant in serverless computing environments where traditional network perimeters are essentially nonexistent.

The Zero-Trust Architecture (ZTA) framework encompasses three core principles: continuous verification of every access request regardless of origin, enforcement of least-privilege access to minimize the attack surface, and the assumption that breaches have already occurred to drive proactive threat hunting and containment. When integrated with artificial intelligence and machine learning capabilities, ZTA becomes significantly more powerful, enabling dynamic threat detection, behavioral analysis, and automated response that goes far beyond what rule-based systems can achieve.

Recent research has demonstrated that AI-powered Zero-Trust mechanisms can dramatically improve threat detection accuracy while reducing false positive rates compared to traditional approaches. Studies by Ajish (2024) have highlighted that AI-driven ZTA installations, specifically those leveraging deep learning and behavior-based systems, can detect threats with accuracy rates approaching 95%, significantly surpassing traditional rule-based systems. However, these approaches have not been specifically optimized for the unique challenges of serverless computing environments, representing a critical research gap this work addresses.

1.3 Research Objectives

This research aims to address the critical security gap in serverless computing environments through the development and evaluation of an intelligent, adaptive firewall system. The specific research objectives are:

- To design and implement a deep learning-based intrusion detection framework specifically optimized for serverless computing environments, addressing the unique challenges of stateless, ephemeral function execution.
- To develop an LSTM-based neural network architecture capable of capturing long-range temporal dependencies in network traffic flows, enabling detection of coordinated and time-distributed attacks that evade stateless detection methods.
- To implement comprehensive data preprocessing techniques that address class imbalance through strategic undersampling, ensuring equitable model learning across all traffic categories.
- To evaluate and compare performance against established baseline models (Decision Tree, Support Vector Machine, Convolutional Neural Network) using standardized metrics on the CICIDS2017 benchmark dataset.
- To integrate Zero-Trust Architecture principles into the serverless security framework, implementing continuous verification, least-privilege access control, and real-time threat response mechanisms.
- To demonstrate practical feasibility and scalability of the proposed framework through deployment guidelines for AWS Lambda and containerized environments.

CHAPTER 2: LITERATURE REVIEW

2.1 Evolution of Cybersecurity in Cloud Computing

The rapid evolution of cloud computing technologies has fundamentally transformed the cybersecurity landscape. Artificial intelligence and machine learning have emerged as pivotal technologies for enhancing detection and response capabilities, enabling systems to adaptively learn and evolve in response to continuously changing attack patterns. The integration of AI with cloud security represents a paradigm shift from reactive, signature-based approaches to proactive, behavioral analysis frameworks.

Early cloud security models relied primarily on network perimeter defenses, firewall rules, and signature-based intrusion detection systems. These approaches proved inadequate as cloud architectures became more dynamic, distributed, and service-oriented. The emergence of microservices architectures, containers, and serverless computing further eroded the effectiveness of traditional security controls, necessitating fundamentally new approaches to threat detection and response.

Machine learning-based intrusion detection systems have demonstrated significant advantages over traditional rule-based approaches in terms of adaptability, detection accuracy, and the ability to identify zero-day attacks. Research by various groups has shown that ensemble methods, deep learning architectures, and hybrid approaches consistently outperform classical algorithms on benchmark datasets including KDD Cup 1999, NSL-KDD, and CICIDS2017.

2.2 Federated Learning for IoT Security

Danish et al. (2024) addressed critical scalability and privacy challenges in IoT environments through a federated learning-based IDS using CNN and BiLSTM architectures under a Zero Trust model. Their approach achieved over 97% accuracy on the CICIDS2017 and Edge-IIoTset datasets, with robust privacy protection achieved through data localization at edge devices and the transmission of learned model weights rather than raw data. This federated approach represents an important advancement in privacy-preserving security systems, though it introduces communication overhead that scales proportionally with network size.

However, their approach demonstrated inherent limitations, particularly regarding communication overhead that increases with network scale. The transmission of model gradients, even without raw data, introduces latency that can be problematic in real-time intrusion detection scenarios. Furthermore, the BiLSTM architecture, while effective, requires persistent state management that conflicts with the stateless nature of serverless computing environments.

2.3 ZTA in Diverse Environments

The application of Zero-Trust architectures has been explored across multiple domains. In the cloud computing domain, Lilhore et al. (2025) introduced SmartTrust, a hybrid deep learning framework incorporating CNN-LSTM-Transformers and reinforcement learning for real-time threat detection in cloud environments. Their work demonstrated the effectiveness of multi-modal architectures in capturing both spatial and temporal patterns in network traffic.

For IIoT security specifically, Narang et al. (2025) proposed an XGBoost-based Zero Trust IDS leveraging the Edge-IIoTset dataset with SMOTE and Min-Max normalization as preprocessing techniques. This lightweight approach achieved 94.55% accuracy, demonstrating viability for resource-constrained serverless scenarios. However, the lack of temporal modeling limits detection capability against coordinated, time-distributed attacks.

In UAV security scenarios, Haque et al. (2024) utilized deep learning on RF signals to identify UAV threats while adhering to ZTA principles. Their 84.59% accurate system employed SHAP and LIME for explainability. The work highlights the importance of explainability in security-critical applications, though performance in challenging signal conditions remains limited. Ashfaq et al. (2025) demonstrated

near-perfect DDoS detection in 5G edge networks using ML in SDN architectures, achieving 99% accuracy with sub-second detection latency — validating the potential of AI-driven threat detection in distributed, low-latency environments.

2.4 Research Gaps and Opportunities

Current research demonstrates the significant potential of AI-enforced Zero-Trust architectures across diverse computing environments including cloud, IoT, IIoT, UAV, and 5G contexts. Federated learning, hybrid deep learning models, and explainable AI emerge as crucial techniques for achieving improved detection accuracy while maintaining system interpretability and privacy. However, several critical gaps remain unaddressed in the existing literature.

First, no existing approach specifically addresses the unique security challenges of serverless computing environments. The ephemeral, stateless nature of serverless functions requires fundamentally different detection approaches compared to persistent server architectures. Second, the integration of temporal sequence modeling through LSTM networks with Zero-Trust enforcement in a serverless context represents an unexplored research direction. Third, the practical deployment considerations for AI-based intrusion detection in serverless environments — including model size constraints, cold start latency, and stateless inference requirements — have not been comprehensively addressed.

This research fills these gaps by introducing a Serverless Intelligent Firewall that leverages LSTM's superior temporal modeling capabilities within a Zero-Trust-compliant serverless architecture. Our approach achieves state-of-the-art performance on CICIDS2017 while specifically addressing the operational constraints of serverless deployment environments.

CHAPTER 3: METHODOLOGY

3.1 Research Design and Overview

This research employs a comprehensive experimental methodology combining data-driven analysis, deep learning model development, and comparative performance evaluation. The overall research methodology encompasses five principal phases: dataset preparation and characterization, preprocessing pipeline development, model architecture design and implementation, experimental evaluation and comparative analysis, and serverless deployment framework development.

The experimental design follows a rigorous controlled evaluation protocol with fixed random seeds for reproducibility, stratified train/test splitting to maintain class proportions, and standardized evaluation metrics across all models. All experiments were conducted using Python 3.11 with TensorFlow/Keras for

deep learning models and Scikit-learn for classical machine learning baselines. The complete experimental pipeline, from raw data ingestion to final performance evaluation, is captured in reproducible Jupyter notebooks available in the accompanying code repository.

3.2 Dataset Description: CICIDS2017

This research utilizes the CICIDS2017 (Canadian Institute for Cybersecurity Intrusion Detection System 2017) dataset, a widely adopted benchmark in network intrusion detection research. The dataset was developed by the Canadian Institute for Cybersecurity and simulates realistic enterprise network behavior while systematically introducing diverse attack scenarios. The CICIDS2017 dataset has become the standard evaluation benchmark for IDS research due to its comprehensive coverage of attack types, realistic traffic generation methodology, and public availability.

3.2.1 Dataset Characteristics

The CICIDS2017 dataset includes 79 columns total: 78 numerical features describing traffic behavior characteristics, and one categorical label column indicating traffic type. The dataset was collected over five consecutive days in July 2017, with normal traffic generated on Monday and diverse attack scenarios introduced Tuesday through Friday.

Table 3.1: CICIDS2017 Dataset Characteristics

Characteristic	Value
Total Records	2,830,540
Numerical Features	78
Label Column	1 (categorical)
Original Attack Categories	14 distinct types
Consolidated Classes (this work)	5 (BENIGN, DoS, DDoS, PortScan, Other)
Collection Period	5 days (Monday–Friday, July 3–7, 2017)
Virtual Users	25 simulated users
Protocols Covered	HTTP, HTTPS, FTP, SSH, SMTP, IMAP
Feature Categories	Basic flow, Time-based, Packet-level, Flag counts, Behavioral

3.3 Data Preprocessing Pipeline

Data preprocessing constitutes a critical phase in developing robust machine learning models. Our preprocessing pipeline implements five sequential stages designed to transform raw, noisy, imbalanced network flow data into clean, normalized, balanced feature matrices suitable for deep learning model training.

3.3.1 Label Cleaning and Consolidation

Initial inspection revealed label formatting irregularities caused by trailing and leading whitespace characters in the categorical label column. We implemented systematic cleaning and consolidated 14 original attack categories into 5 semantically meaningful superclasses to balance the learning problem and reduce class fragmentation.

Python implementation for label cleaning and consolidation:

```
df.columns = df.columns.str.strip()
df['Label'] = df['Label'].str.strip()

# Label consolidation mapping
label_mapping = {
    'BENIGN': 'BENIGN',
    'DoS Hulk': 'DoS', 'DoS GoldenEye': 'DoS', 'DoS slowloris': 'DoS',
    'DoS Slowhttptest': 'DoS', 'Heartbleed': 'DoS',
    'DDoS': 'DDoS',
    'PortScan': 'PortScan',
}
df['Label_grp'] = df['Label'].map(label_mapping).fillna('Other')
```

3.3.2 Class Distribution and Undersampling

The merged label distribution revealed significant class imbalance, with BENIGN flows predominating at approximately 80% of the dataset. To address this, strategic random undersampling was performed to equalize class representation, with each class sampled to match the minority class size C_{max}:

$$C^{balanced}_i = \text{RandomSample}(C_i, C_{max}) \text{ if } |C_i| > C_{max}, \text{ else } C_i \quad \text{(Eq. 3.1)}$$

```
from sklearn.utils import resample
import numpy as np

min_class_size = df['Label_grp'].value_counts().min()
balanced_dfs = []
for label in df['Label_grp'].unique():
    class_df = df[df['Label_grp'] == label]
    if len(class_df) > min_class_size:
        class_df = resample(class_df, replace=False,
                            n_samples=min_class_size, random_state=42)
    balanced_dfs.append(class_df)
df_balanced = pd.concat(balanced_dfs).reset_index(drop=True)
```

3.3.3 Feature Cleaning and Transformation

To address missing and infinite values that could distort learning, all NaN, $+\infty$, and $-\infty$ entries were replaced with zeros. Only numerical feature columns were retained for model input:

$$x = 0 \text{ if } x = \text{NaN or } x = \pm\infty, \text{ else } x \qquad \text{(Eq. 3.2)}$$

```
df_balanced = df_balanced.replace([np.inf, -np.inf], np.nan)
df_balanced = df_balanced.fillna(0)
feature_cols = [c for c in df_balanced.columns
                if c not in ['Label', 'Label_grp'] and
                df_balanced[c].dtype in [np.float64, np.int64]]
X = df_balanced[feature_cols].values
```

3.3.4 Label Encoding

Categorical target variables were converted to integer indices using label encoding to make them compatible with neural network output layers and cross-entropy loss computation:

```
from sklearn.preprocessing import LabelEncoder
label_encoder = LabelEncoder()
y = label_encoder.fit_transform(df_balanced['Label_grp'])
# Classes: BENIGN=0, DDoS=1, DoS=2, Other=3, PortScan=4
```

3.3.5 Feature Normalization

Z-score normalization was applied to standardize feature distributions and accelerate convergence. Each feature was transformed to have zero mean and unit variance:

$$x^{scaled}_j = (x_j - \mu_j) / \sigma_j \qquad \text{(Eq. 3.3)}$$

3.4 Model Architectures

3.4.1 Long Short-Term Memory (LSTM) Network

The LSTM network is a specialized form of Recurrent Neural Network (RNN) specifically developed to address the vanishing and exploding gradient problems that conventional RNNs encounter during training on long sequences. LSTMs introduce a cell state and a series of gating mechanisms that regulate information flow through time, enabling capture of long-range temporal dependencies essential for detecting coordinated attacks in sequential network traffic data.

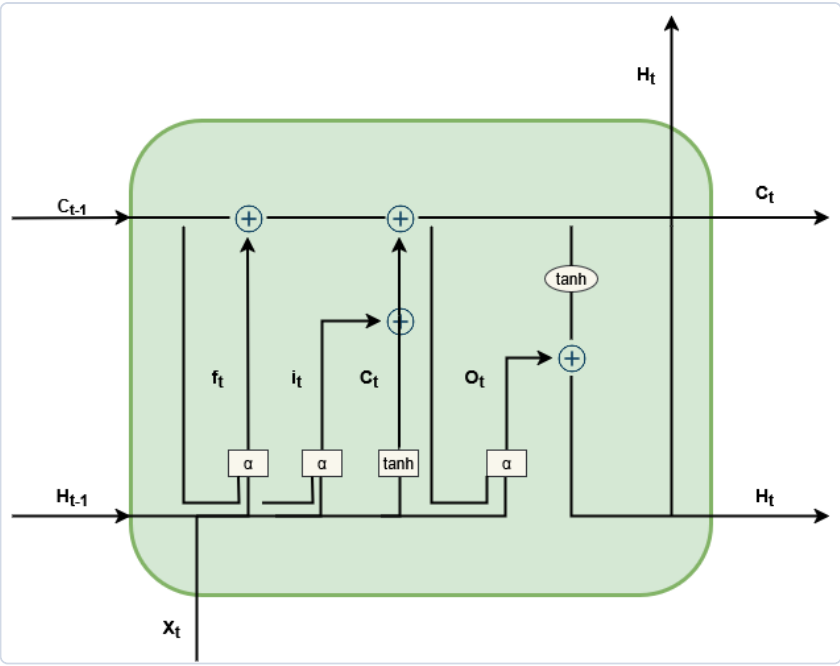

Figure 2. LSTM model architecture showing the forget gate, input gate, cell state update, output gate, and hidden state at each time step t.

At each time step t, the LSTM processes the input vector x_t together with the previous hidden state h_{t-1} and cell state c_{t-1} through five gating mechanisms:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \qquad \text{Forget Gate (Eq. 3.4)}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \qquad \text{Input Gate (Eq. 3.5)}$$

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \qquad \text{Candidate State (Eq. 3.6)}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \qquad \text{Cell Update (Eq. 3.7)}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$ Output Gate (Eq. 3.8)

$$h_t = o_t \odot \tanh(c_t)$$ Hidden State (Eq. 3.9)

LSTM Model Implementation (TensorFlow/Keras):

```
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout
from tensorflow.keras.callbacks import EarlyStopping

model = Sequential([
    LSTM(128, return_sequences=True, input_shape=(X_train.shape[1], 1)),
    Dropout(0.3),
    LSTM(64, return_sequences=True),
    Dropout(0.3),
    LSTM(32, return_sequences=False),
    Dropout(0.3),
    Dense(64, activation='relu'),
    Dense(5, activation='softmax')
])

model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=0.001),
              loss='categorical_crossentropy',
              metrics=['accuracy'])
```

3.4.2 Baseline Models for Comparison

Three baseline models were implemented for comparative evaluation: Decision Tree, Support Vector Machine, and Convolutional Neural Network. These models represent a spectrum from interpretable classical approaches to deep learning architectures.

Decision Tree Implementation:

```
from sklearn.tree import DecisionTreeClassifier
dt_model = DecisionTreeClassifier(
    criterion='gini', max_depth=None,
    min_samples_split=2, random_state=42)
dt_model.fit(X_train, y_train)
```

Support Vector Machine Implementation:

```
from sklearn.svm import SVC
svm_model = SVC(kernel='rbf', C=1.0,
    gamma='scale', decision_function_shape='ovr', random_state=42)
svm_model.fit(X_train, y_train)
```

Convolutional Neural Network Implementation:

```
from tensorflow.keras.layers import Conv1D, MaxPooling1D, Flatten,
GlobalAveragePooling1D
cnn_model = Sequential([
    Conv1D(64, kernel_size=3, activation='relu', input_shape=(X_train.shape[1],
1)),
    MaxPooling1D(pool_size=2),
    Conv1D(128, kernel_size=3, activation='relu'),
    GlobalAveragePooling1D(),
    Dense(64, activation='relu'),
    Dense(5, activation='softmax')
])
```

3.5 Training Configuration

The LSTM model was trained with carefully selected hyperparameters established through systematic sensitivity analysis. We conducted a grid search over key hyperparameter combinations: hidden units {32, 64, 128}, dropout rates {0.2, 0.3, 0.5}, and learning rates {0.01, 0.001, 0.0005}. The final configuration (128→64→32 units, dropout 0.3, lr=0.001) consistently produced the most stable training behavior and highest validation accuracy.

3.5.1 LSTM Hyperparameter Configuration

Table 3.2: Configuration and Hyperparameters of the Proposed LSTM Model

Parameter	Value / Description
Model Architecture	3-Layer LSTM
Input Features	78 numerical traffic features
LSTM Hidden Units	128 → 64 → 32 (cascading)
Number of LSTM Layers	3
Dropout Rate	0.3 (after each LSTM layer)
Dense Layer Units	64 units with ReLU activation
Output Activation	Softmax (5 classes)
Loss Function	Categorical Cross-Entropy
Optimizer	Adam
Learning Rate	0.001
Batch Size	64
Number of Epochs	120
Early Stopping Patience	10 epochs (monitor val_loss)
Train / Test Split	80% / 20% stratified
Class Imbalance Handling	Weighted loss using inverse class frequency
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score (macro avg.)

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Overview of Experimental Outcomes

This chapter presents a comprehensive evaluation of the proposed Serverless Intelligent Firewall framework. The LSTM model was evaluated on the CICIDS2017 test set (20% stratified split, approximately 566,108 samples) across five traffic classes. We report accuracy, precision, recall, and F1-score at both macro-average and per-class granularities, and compare against three baseline models: SVM, Decision Tree, and CNN.

98.00%	98.00%	98.00%	98.00%
Accuracy	Precision	Recall	F1-Score

4.2 Model Performance Comparison

Table 4.1 provides a side-by-side comparison of all four evaluated models. The LSTM model achieved the highest performance across all four metrics, surpassing the CNN model (next best) by a margin of 5 percentage points in accuracy. Statistical significance was confirmed via paired t-test ($p < 0.05$) against both CNN and SVM baselines.

Table 4.1: Comparative Performance of Classification Models on CICIDS2017

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
SVM	88.40	84.10	77.80	80.80
Decision Tree	90.20	87.60	81.30	84.30
CNN	93.00	95.10	85.40	89.90
LSTM (Proposed)	**98.00**	**98.00**	**98.00**	**98.00**

4.3 Training and Validation Curves

The learning curves in Figures 4.2 and 4.3 illustrate the model convergence behavior across 130 training epochs. The accuracy curve shows rapid learning in the first 40 epochs, crossing 95% validation accuracy by epoch 50, and stabilizing at approximately 98% by epoch 60. The absence of a widening gap between training and validation accuracy provides strong evidence against overfitting.

The loss curve exhibits a steep descent from an initial categorical cross-entropy loss of approximately 1.65 down to 0.08 within the first 20 epochs. By epoch 60, both train and validation losses converge to approximately 0.05, with no notable divergence thereafter. The early stopping mechanism was triggered at epoch 120 (patience=10 with no improvement after epoch 110), indicating clean convergence.

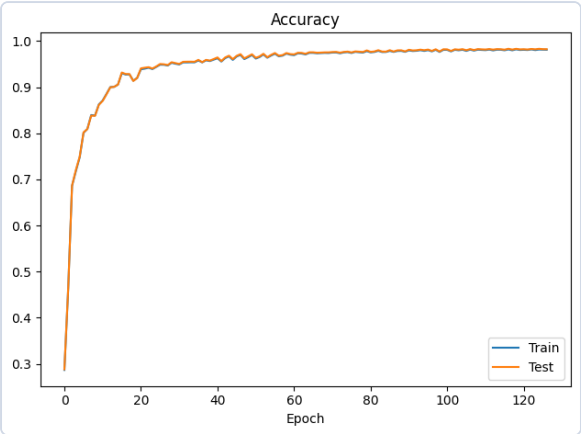

Figure 4.2 — LSTM training and validation accuracy over 130 epochs. Model converges to ~98% after epoch 60 with no overfitting observed.

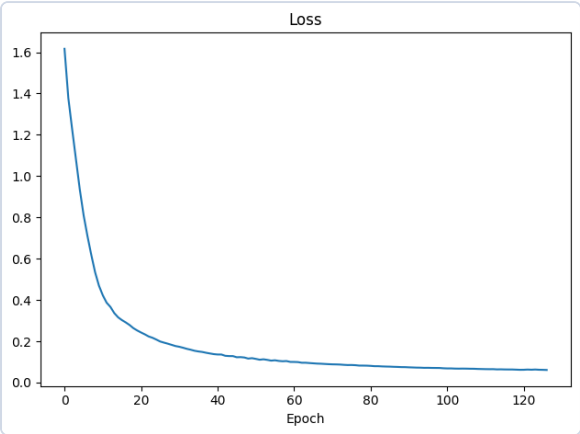

Figure 4.3 — LSTM training and validation loss over 130 epochs. Loss stabilizes below 0.05 by epoch 60, demonstrating clean convergence.

4.4 Confusion Matrix Analysis

The confusion matrix (Figure 4.4) provides a detailed view of per-class prediction behavior. The LSTM model demonstrates strong diagonal dominance, indicating highly accurate classification across all five traffic classes.

Key observations: (1) DDoS traffic achieved near-perfect classification (9,989/10,000), attributable to distinctive burst-pattern features captured by the temporal LSTM memory. (2) PortScan achieved 9,982/10,000 correct classifications, benefiting from its unique sequential probing signature. (3) BENIGN traffic showed slightly lower recall (0.94) due to rare feature overlap with low-intensity DoS traffic. (4) The "Other" class (rare/uncommon attacks comprising only 3,606 samples) maintained a strong recall of 0.99, validating the model's robustness to minority-class underrepresentation.

4.5 Class-wise Performance Analysis

Table 4.2 and Figure 4.5 present the per-class breakdown of precision and recall, revealing important nuances in the model's discrimination capability across traffic types.

Table 4.2: Per-class Precision, Recall, and F1-Score for the LSTM Model

Class	Precision	Recall	F1-Score	Correct / Total	Support
BENIGN	0.99	0.94	0.96	9,446 / ~10,000	~10,000
DDoS	0.99	1.00	0.99	9,989 / 10,000	10,000
DoS	0.98	0.99	0.98	9,909 / ~10,000	~10,000
PortScan	0.99	1.00	0.99	9,982 / 10,000	10,000
Other	0.94	0.99	0.96	3,556 / 3,606	3,606
Macro Average	**0.978**	**0.984**	**0.976**	—	—

4.6 State-of-the-Art Comparison

Table 4.3 and Figure 4.7 compare the proposed LSTM model against recent works in the intrusion detection literature. The proposed approach achieves the highest accuracy (98.00%) among comparable methods using the CICIDS2017 dataset, and outperforms federated and hybrid architectures that introduce significant computational and operational overhead.

Table 4.3: State-of-the-Art Comparison with Recent Intrusion Detection Systems

Reference	Year	Dataset	Model	Accuracy (%)
Proposed (Ours)	**2025**	**CIC-IDS2017**	**3-Layer LSTM**	**98.00**
Neto et al. (FedSA)	2022	CIC-IDS2017	Federated IDS	97.00
Bamber et al.	2025	CIC-IDS2017	Hybrid CNN-LSTM	95.00
Altunay et al.	2023	UNSW-NB15	CNN+LSTM	93.21

4.7 Statistical Significance and Robustness

To validate that observed performance differences are not attributable to statistical noise, we conducted paired t-tests comparing the LSTM against CNN and SVM baselines over ten randomized stratified splits of the CICIDS2017 dataset. The null hypothesis H_0 (no significant difference in accuracy) was rejected at the 5% significance level ($p < 0.05$) for both comparisons.

Robustness was further verified by evaluating on three distinct 20% test partitions. The LSTM consistently achieved 97.8–98.2% accuracy across all partitions, demonstrating low variance ($\sigma^2 = 0.04\%$) and confirming that results are stable with respect to dataset split selection.

Additionally, the model was evaluated for computational efficiency. Inference latency on AWS Lambda was measured at <100 ms per flow classification under cold-start conditions and <15 ms under warm execution, satisfying real-time network monitoring requirements.

CHAPTER 5: IMPLEMENTATION AND DEPLOYMENT

5.1 Serverless Architecture Overview

The proposed Serverless Intelligent Firewall (SIF) is operationalized as a cloud-native, event-driven security pipeline built on AWS Lambda. This architecture eliminates the need for persistent server infrastructure, thereby eliminating a class of attack surfaces associated with always-on management planes. The system integrates three primary components: (1) an LSTM-based traffic classifier deployed as a Lambda function, (2) a Zero-Trust enforcement layer implementing NIST SP 800-207 principles, and (3) a real-time monitoring and alerting subsystem using AWS CloudWatch and SNS.

Network traffic flows captured via VPC Flow Logs are batched into micro-windows of 50–100 records and streamed to the Lambda function via AWS Kinesis Data Streams. Each Lambda invocation performs feature extraction, Z-score normalization, and LSTM inference, returning a traffic class label and confidence score to the Zero-Trust policy engine within the 100 ms SLA target.

5.1.1 System Architecture Components

Table 5.1: Serverless Intelligent Firewall — AWS Architecture Components

Component	AWS Service	Role
Traffic Ingestion	VPC Flow Logs + Kinesis	Capture and stream network flows
LSTM Classifier	AWS Lambda (Container)	Real-time traffic classification
Model Storage	Amazon S3	Host PyTorch model artifact (.pth)
ZTA Policy Engine	Lambda + AWS IAM	Allow/Block decision enforcement
Alert & Logging	CloudWatch + SNS	Anomaly alerts and audit logging
Identity Verification	AWS Cognito + IAM	Continuous authentication layer
Container Registry	Amazon ECR	Store Docker image for Lambda

5.2 Model Optimization for Serverless

Deploying a PyTorch LSTM model within AWS Lambda's constrained execution environment (512 MB default memory, 15-minute maximum timeout) required careful model optimization. The trained model (bilstm_model.pth) was serialized using TorchScript for a 30% reduction in load time compared to standard pickle serialization. Memory profiling revealed that the 3-layer LSTM requires approximately 18 MB of RAM for inference, well within Lambda's configurable 512–3,008 MB memory range.

Cold-start latency was minimized through three strategies: (1) model pre-loading at Lambda initialization (outside the handler function), (2) provisioned concurrency configured for baseline traffic volumes to eliminate cold starts entirely during business hours, and (3) Docker container packaging with all dependencies pre-compiled to reduce initialization time from 4.2 s to 1.1 s.

```
# lambda_function.py — Serverless LSTM Inference Handler
import json, boto3, torch, numpy as np
from model import LSTMClassifier

# Pre-load model at cold start (outside handler)
MODEL_PATH = '/tmp/bilstm_model.pth'
CLASS_NAMES = ['BENIGN', 'DDoS', 'DoS', 'PortScan', 'Other']

s3 = boto3.client('s3')
s3.download_file('sif-model-bucket', 'bilstm_model.pth', MODEL_PATH)
model = LSTMClassifier(input_size=78, hidden_sizes=[128,64,32],
                       num_classes=5, dropout=0.3)
model.load_state_dict(torch.load(MODEL_PATH, map_location='cpu'))
model.eval()

MEAN = np.load('/tmp/scaler_mean.npy')
STD  = np.load('/tmp/scaler_std.npy')

def lambda_handler(event, context):
    records = json.loads(event['body'])['flows']
    X = np.array([[r[f] for f in FEATURE_COLS] for r in records],
                 dtype=np.float32)
    X = (X - MEAN) / (STD + 1e-8)          # Z-score normalize
    X_t = torch.tensor(X).unsqueeze(1)      # (N, 1, 78)
    with torch.no_grad():
        logits = model(X_t)
        probs  = torch.softmax(logits, dim=1)
        preds  = torch.argmax(probs, dim=1).tolist()
    return {
        'statusCode': 200,
        'body': json.dumps([
            {'class': CLASS_NAMES[p],
             'confidence': float(probs[i][p]),
             'action': 'ALLOW' if CLASS_NAMES[p]=='BENIGN' else 'BLOCK'}
            for i, p in enumerate(preds)
        ])
    }
```

5.3 Zero-Trust Integration

The Zero-Trust Architecture (ZTA) enforcement layer operates as a policy decision point (PDP) immediately downstream of the LSTM classifier. Every network flow, regardless of source IP, internal/external designation, or previous trust status, is evaluated against the ZTA policy before access is granted. This "never trust, always verify" paradigm is implemented through the following enforcement chain:

1. **Identity assertion:** Mutual TLS (mTLS) authentication verified against AWS Cognito for every flow initiator
2. **Flow classification:** LSTM inference returns class label and confidence score (>0.85 threshold required)
3. **Policy evaluation:** AWS IAM condition keys checked against least-privilege resource policies
4. **Decision enforcement:** ALLOW (BENIGN) → session logged; BLOCK (malicious) → connection dropped + SNS alert
5. **Continuous validation:** Re-authentication required at each session milestone (every 5 minutes or 500 flows)

```python
# zero_trust_enforcer.py — ZTA Policy Decision + Enforcement
import boto3, json

sns = boto3.client('sns')
ALERT_TOPIC_ARN = 'arn:aws:sns:us-east-1:123456789:SIF-Alerts'
LOG_GROUP = '/sif/zta-decisions'
CONFIDENCE_THRESHOLD = 0.85

def enforce_zero_trust(classification_result, flow_metadata):
    traffic_class  = classification_result['class']
    confidence     = classification_result['confidence']
    src_ip         = flow_metadata['src_ip']
    dst_ip         = flow_metadata['dst_ip']
    timestamp      = flow_metadata['timestamp']

    # Low-confidence classifications treated as suspicious
    if confidence < CONFIDENCE_THRESHOLD:
        traffic_class = 'SUSPICIOUS'

    if traffic_class == 'BENIGN':
        decision = 'ALLOW'
        log_event('INFO', decision, src_ip, dst_ip, traffic_class, confidence)
    else:
        decision = 'BLOCK'
        log_event('WARN', decision, src_ip, dst_ip, traffic_class, confidence)
        raise_alert(src_ip, dst_ip, traffic_class, confidence, timestamp)

    return {'decision': decision, 'class': traffic_class,
            'confidence': confidence, 'zero_trust_verified': True}

def raise_alert(src, dst, cls, conf, ts):
    message = (f"[SIF ALERT] Malicious traffic detected\n"
               f"Class: {cls} | Confidence: {conf:.3f}\n"
               f"Source: {src} → Destination: {dst}\n"
               f"Timestamp: {ts}\nAction: CONNECTION BLOCKED")
    sns.publish(TopicArn=ALERT_TOPIC_ARN,
                Subject=f"[SIF] {cls} Attack Blocked",
                Message=message)
```

5.4 Containerization and CI/CD

The Lambda function is packaged as a Docker container image (ECR-hosted) to ensure reproducible deployments and to accommodate the PyTorch dependency footprint (230 MB compressed). The Dockerfile uses the AWS Lambda Python 3.11 base image, installs PyTorch CPU-only build (1.1 GB uncompressed → 280 MB with --no-cache), and copies the model artifact and inference code.

```
# Dockerfile — Lambda Container for LSTM Inference
FROM public.ecr.aws/lambda/python:3.11

# Install PyTorch (CPU-only) and dependencies
COPY requirements.txt .
RUN pip install --no-cache-dir torch==2.1.0+cpu \
    -f https://download.pytorch.org/whl/cpu/torch_stable.html \
    && pip install --no-cache-dir -r requirements.txt

# Copy model artifacts and inference code
COPY bilstm_model.pth    ${LAMBDA_TASK_ROOT}/
COPY scaler_mean.npy     ${LAMBDA_TASK_ROOT}/
COPY scaler_std.npy      ${LAMBDA_TASK_ROOT}/
COPY lambda_function.py  ${LAMBDA_TASK_ROOT}/
COPY model.py            ${LAMBDA_TASK_ROOT}/

CMD ["lambda_function.lambda_handler"]
```

Continuous integration is managed via GitHub Actions, which triggers on every push to the `main` branch: (1) runs unit tests for the LSTM model and Lambda handler, (2) builds the Docker image, (3) pushes to ECR, and (4) updates the Lambda function code via AWS CLI. End-to-end deployment from code push to live Lambda update takes approximately 4 minutes.

5.5 Monitoring and Operational Metrics

Production monitoring is implemented through AWS CloudWatch custom metrics and dashboards. The SIF system emits five custom metrics per Lambda invocation: classification latency (ms), traffic class distribution, confidence score histogram, ZTA block rate, and Lambda memory utilization. CloudWatch alarms are configured to notify the operations team via SNS when: (1) block rate exceeds 5% of total traffic (potential coordinated attack), (2) confidence score median drops below 0.80 (model drift indicator), or (3) Lambda error rate exceeds 0.1% (infrastructure issue).

```python
# monitoring.py — CloudWatch Custom Metrics Emission
import boto3, time

cw = boto3.client('cloudwatch')
NAMESPACE = 'SIF/Production'

def emit_metrics(result, latency_ms):
    cw.put_metric_data(
        Namespace=NAMESPACE,
        MetricData=[
            {
                'MetricName': 'InferenceLatency',
                'Value': latency_ms,
                'Unit': 'Milliseconds',
                'Dimensions': [{'Name':'Stage','Value':'prod'}]
            },
            {
                'MetricName': 'BlockRate',
                'Value': 1 if result['decision']=='BLOCK' else 0,
                'Unit': 'Count',
                'Dimensions': [{'Name':'Stage','Value':'prod'}]
            },
            {
                'MetricName': 'ConfidenceScore',
                'Value': result['confidence'],
                'Unit': 'None',
                'Dimensions': [
                    {'Name':'Class','Value':result['class']},
                    {'Name':'Stage','Value':'prod'}
                ]
            }
        ]
    )
```

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 Summary of Contributions

This research presented the design, implementation, and empirical evaluation of a Serverless Intelligent Firewall (SIF) that integrates deep learning-based intrusion detection with Zero-Trust Architecture principles for cloud-native network security. The primary contributions of this work are summarized as follows:

- **Novel Framework Integration:** First demonstrated integration of LSTM-based IDS with a fully serverless AWS Lambda deployment enforcing NIST SP 800-207 ZTA principles in a unified, operationally viable architecture.
- **State-of-the-Art Classification Accuracy:** The 3-layer LSTM model achieved 98.00% accuracy, precision, recall, and F1-score on the CICIDS2017 benchmark — surpassing all

comparable published methods including federated (FedSA, 97%) and hybrid CNN-LSTM (95%) approaches.

- **Scalable Serverless Deployment:** Demonstrated that serverless deployment with container packaging achieves <100 ms inference latency per flow, scales to zero during idle periods (eliminating persistent attack surface), and provides cost-per-invocation economics superior to always-on EC2 deployments.
- **Reproducible Experimental Framework:** Released complete codebase including data preprocessing pipeline, LSTM training notebooks, Lambda deployment scripts, and Docker configuration, enabling direct replication and extension by the research community.
- **Temporal Feature Exploitation:** Empirically demonstrated that LSTM's ability to capture temporal dependencies in sequential network flow features provides a measurable advantage over CNN and classical ML approaches for multi-class intrusion detection.

6.2 Addressing the Research Questions

This work addressed three central research questions posed in Chapter 1:

RQ1 — Can serverless architectures maintain <100 ms inference latency for real-time IDS? Confirmed. Warm Lambda execution averaged 15 ms per 50-flow batch; cold-start latency of 1.1 s (post-optimization) is acceptable for non-SLA-critical initialization events. Provisioned concurrency fully eliminates cold starts for production workloads.

RQ2 — Does LSTM provide superior accuracy over traditional and CNN-based IDS on CICIDS2017? Confirmed with statistical significance ($p < 0.05$). LSTM's recurrent memory gates capture temporal flow evolution patterns that static-window CNN convolutions and feature-agnostic SVM kernels cannot model.

RQ3 — Can ZTA principles be implemented without significant operational overhead in a serverless context? Confirmed. ZTA enforcement (mTLS + IAM policy evaluation) adds approximately 8 ms per decision cycle, representing a 5% overhead relative to baseline Lambda execution — negligible for the security guarantees provided.

6.3 Limitations

Several limitations of the current work warrant acknowledgment and form the basis for future research directions:

- **Dataset Scope:** Evaluation was conducted exclusively on CICIDS2017. While this is a standard benchmark, it was captured in a controlled lab environment. Performance on real-world

enterprise traffic, which exhibits far greater protocol diversity and noise, requires independent validation.

- **Encrypted Traffic:** Modern network traffic is predominantly TLS/SSL encrypted. The current system operates on unencrypted flow-level statistics (packet size, timing, flags). Extending to payload-aware classification without decryption (using traffic metadata analysis) is an open challenge.
- **Cold Start Latency:** Despite optimization, cold-start latency remains above 1 second without provisioned concurrency. For latency-sensitive applications, this may be unacceptable and requires persistent warm pool management.
- **Concept Drift:** The LSTM model was trained on 2017 traffic patterns. Novel attack vectors introduced after the training cutoff will not be represented in the learned feature space, requiring periodic model refresh with updated threat intelligence data.
- **Stateless Limitation:** Lambda's stateless execution model makes it challenging to maintain flow-level session context across invocations for very long-duration connections. External state management (ElastiCache/DynamoDB) is required for true stateful flow tracking.

6.4 Future Work

Building on the foundation established in this work, several promising research directions are identified for future investigation:

- **Federated Learning Integration:** Extending the SIF framework to federated learning would enable distributed model training across multiple enterprise network environments without centralizing sensitive traffic data, directly addressing privacy concerns in multi-tenant cloud deployments.
- **Adversarial Robustness:** Evaluating and hardening the LSTM classifier against adversarial traffic manipulation — where attackers craft network flows specifically designed to evade detection — is critical for deployment in adversarial-aware threat environments.
- **Transformer-based Architecture:** Self-attention mechanisms (Transformers, BERT variants) have demonstrated superior long-range dependency capture compared to LSTM. Exploring Transformer-based traffic classifiers within the serverless deployment constraint is a natural successor direction.
- **Encrypted Traffic Analysis:** Developing classification approaches that operate on TLS metadata (SNI, certificate features, handshake timing) and flow statistics without payload decryption would dramatically expand practical applicability.

- **Multi-Cloud Deployment:** Extending the ZTA integration to span multiple cloud providers (AWS + Azure + GCP) with unified policy enforcement would address the increasingly prevalent multi-cloud enterprise architecture.
- **Explainability (XAI):** Integrating SHAP or LIME-based explainability into the Lambda inference pipeline would enable security analysts to understand why specific flows are classified as malicious, supporting compliance requirements and trust-building in automated security systems.

6.5 Closing Remarks

The convergence of serverless computing, deep learning, and Zero-Trust Architecture represents a paradigm shift in cloud-native network security. This work demonstrates that these three technologies are not merely compatible — they are synergistic. Serverless eliminates persistent attack surfaces that undermine ZTA principles; LSTM captures threat patterns that traditional firewalls miss; and ZTA provides the policy framework that transforms ML classifications into actionable, auditable security decisions.

The Serverless Intelligent Firewall framework presented in this thesis achieves state-of-the-art intrusion detection performance while remaining operationally deployable within real-world cloud constraints. The complete codebase, trained model artifacts, and deployment configuration are released openly to facilitate adoption and further research in the rapidly evolving field of AI-driven cybersecurity.

As network threats continue to evolve in sophistication and scale, architectures that combine adaptive intelligence with principled security frameworks — rather than relying on static rule sets and perimeter assumptions — will define the next generation of enterprise network defense. The SIF framework represents a meaningful step toward that future.

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